DONGSU KIM

+82 10-9067-5828 | kumdingso@gmail.com | Artificial Intelligence & Media Lab. (AIMLab)

Education

Hanbat National University (Advisor: Prof. Haneol Jang) *Daejeon*

M.S. Student in Computer Engineering Mar. 2025 - Present

GPA 4.43/4.5

Hanbat National University (Advisor: Prof. Haneol Jang) *Daejeon*

B.S. in Computer Engineering Mar. 2019 - Feb. 2025

GPA 3.96/4.5 (Major: 4.21/4.5)

Publications

Conference

[1] **Dongsu Kim**, Seoyeon Oh, Cheoneum Park, and Haneol Jang. "Dual-Stream Dual-Backbone Architecture for HyperSpectral Face Anti-Spoofing", *The 17th International Conference on ICT Convergence*, Oct. 2024.

[2] Dabin Kim, Seoyeon Oh, **Dongsu Kim**, Cheoneum Park, and Haneol Jang. "Face anti-spoofing method using contrastive learning and color jittering", *2024 Korean Digital Forensic Society Winter Conference*, Dec. 2024.

[3] Hyeongwon Seo, Jisang Lee, **Dongsu Kim**, Seoyeon Oh, and Haneol Jang. "Effective Detection of Generated Images Using Fast Fourier Transform and Discrete Cosine Transform", *2023 Korean Digital Forensic Society Winter Conference*, Dec. 2023.

[4] **Dongsu Kim**, Jisang Lee, Seoyeon Oh, and Haneol Jang. "Effective Data Augmentaions for Fingerprint Recognition", *2023 Korean Digital Forensic Society Summer Conference*, Jun. 2023.

[5] Jisang Lee, **Dongsu Kim**, Seoyeon Oh, and Haneol Jang. "Lightweight Fingerprint Forgery Detection Technique Utilizing Median Filterand Ensemble Model", *2023 Korean Digital Forensic Society Summer Conference*, Jun. 2023.

[6] Seoyeon Oh, Jisang Lee, **Dongsu Kim**, and Haneol Jang. "Lightweight Deep Convolutional Neural Networks for Fine-Grained Leaf Classification", *The 33rd Joint Conference on Communications and Information*, Apr. 2023.

[7] Eojin Jang, Jisang Lee, **Dongsu Kim**, Seoyeon Oh, and Haneol Jang. "Development of Optical Character Recognition Model for Document Leak Investigation", *2022 Korean Digital Forensic Society Winter Conference*, Dec. 2022.

Journal

[1] **Dongsu Kim**, Jincheol Ahn, Youngjoon Yoo, Taewoo Oh, and Haneol Jang. **"JPEG-aware steganalysis: explicit frequency modeling with dequantized DCT for real-world JPEG images"**, ***Scientific Reports***, *vol. 16, article 30309*, 2026. **(SCIE, IF 4.9, JCR Top 14.6%)**

[2] **Dongsu Kim**, Seoyeon Oh , Haecheol Choi, Taewoo Oh, and Haneol Jang. "Image Distortion-Based Attention for JPEG Compressed-Domain Steganalysis", *Korean Digital Forensic Society, vol. 19, no. 4*, 2025.

[3] Dabin Kim, Seoyeon Oh, **Dongsu Kim**, Cheoneum Park, and Haneol Jang. "Face anti-spoofing method using contrastive learning and color jittering", *Journal of Digital Forensics, vol. 18, no. 5*, 2024.

[4] **Dongsu Kim**, Seoyeon Oh , Jisang Lee, and Haneol Jang. "Effective Data Augmentations for Cross-Domain Fingerprint Recognition", *Korean Digital Forensic Society, vol. 18, no. 1*, 2024.

Research Projects

Development of LLM-Based Text Steganography with Enhanced Resistance to Steganalysis *Python, PyTorch*

Independent Research Jan. 2026 - Present

- Research on information hiding in LLM-generated text
- Development of text steganography methods focused on resistance to steganalysis

Research on Generative AI-Based Image Watermarking Technologies *Python, PyTorch*

National Security Research Institute Mar. 2026 - Present

- Data analysis on watermarking
- Development of **generative AI-based watermarking model**
- Development of **generative AI-based watermarking algorithm**

Development of a Real-World JPEG Steganalysis Framework *Python, PyTorch*

Independent Research Apr. 2025 - Jan. 2026

- Construction of a real-world JPEG steganalysis benchmark (RWSteg)
- Development of a JPEG-aware steganalysis model using dequantized DCT coefficients
- Development of ROI extraction for arbitrary-resolution JPEG images
- Analysis of quantization table and chroma subsampling mismatches

Research on ROI Extraction for Steganalysis and Visualization of Hidden-Region Inference *Python, PyTorch*

National Security Research Institute Apr. 2025 - Oct. 2025

- Data analysis on steganography
- Development of **visualization algorithm** for steganalysis model
- Development of **ROI extraction algorithm** for steganalysis

Development of High Accuracy Steganalysis Model *Python, PyTorch*

Capstone Design Jan. 2024 - Sep. 2024

- Improvement of steganalysis performance through adversarial training

Research on Solutions to Challenges in Real-World Image Steganalysis *Python, PyTorch, Matlab*

National Security Research Institute Apr. 2023 - Oct. 2024

- Construct of a **real-world mobile image steganography dataset**
- Development of **robust steganalysis** under **CSM(cover-source mismatch)** conditions

Awards and Honors

Best Paper Award	President's Award(학회장상), *2024 Korean Digital Forensic Society Winter Conference*	*December, 2024*
Silver Award	*13th College of Information Technology Project Exhibition*, Hanbat National University	*November, 2024*
Competition Award	President's Award(소프트웨어중심대학협의회장상), *2023 SW-Centered University AI/SW Competition*	*August, 2023*
Competition Award	President's Award(소프트웨어중심대학협의회장상), *2022 SW-Centered University AI/SW Competition*	*November, 2022*

Technical Skills

Programming	Python, Java, C, C++
Professional Softwares	PyTorch, Matlab
Languages	Korean(Native), English(Intermediate)

Teaching Experiences

Computer Vision Class *Hanbat National University*

Teaching Assistant (Prof. Haneol Jang) Sep. 2025 - Dec. 2025

Artifical Intelligence Class *Hanbat National University*

Teaching Assistant (Prof. Haneol Jang) Mar. 2025 - June 2025

Artificial Intelligence and Machine Learning Class *Hanbat National University*

Teaching Assistant (Prof. Haneol Jang) Mar. 2024 - June 2024

Artificial Intelligence and Applications Class *Hanbat National University*

Teaching Assistant (Prof. Haneol Jang) Mar. 2024 - June 2024

Data Structure Class *Hanbat National University*

Teaching Assistant (Prof. Haneol Jang) Mar. 2023 - June 2023